GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: THE CAPACITY-SUBSTITUTION CHANNEL AND A DISTRIBUTION-FREE NET-CARBON-REDUCTION CONDITION

TEAM A1 — ROLE 1 (GENERAL CARBON THEORY)

ABSTRACT. We harden and generalize the carbon analysis of the flexible-computing-load (FCL) model beyond the uniform-distribution benchmark. Three results. *First*, we re-derive the uniform closed forms (paper eqs. 18–26) from the first-order conditions and verify them to machine precision against direct quadrature; along the way we establish the *knife-edge* structure for the entire power-law family and reconcile two transcription issues in the current draft. *Second*, we prove the central comparative-statics claim $K_R^* > K_R^\circ$—renewable capacity is strictly higher with the FCL program—for *any* residual-demand distribution satisfying a single, economically transparent condition on the *shift ratio* $g(x) = F_{K_R}(x)/f(x)$: the program raises renewable capacity whenever an extra megawatt of renewables relaxes residual demand more in deep-surplus states than in tight states. The power-law family satisfies this for every shape parameter, which is exactly why footnote 6's extension goes through. *Third*, we give the exact net-carbon inequality, decompose it into a distribution-free *utilization* effect and a sign-controlled *capacity-substitution* effect, and state a clean sufficient condition for the carbon part of the triple win. Every quantitative claim is checked numerically. A companion gap memo records precisely where the fully general argument resists and what would close it.

1. Setup and notation

We work with the model of Section 3 of the paper and adopt its notation. A regulated utility chooses renewable capacity K_R, dispatchable fossil backup K_F, and flexible-computing-load capacity K_M to maximize expected profit over one representative period. The state variable is *residual demand*

$$\tilde{x} := \tilde{D} - K_R \tilde{S} \in [-K_R, \bar{D}], \tag{1}$$

where $\tilde{D} \in [0, \bar{D}]$ is stochastic consumer demand and $\tilde{S} \in [0,1]$ the renewable capacity factor. Let $F(x; K_R)$ be the CDF of $\tilde{x}$ with density $f(x; K_R)$, extended by $F \equiv 0$ for $x < -K_R$ and $F \equiv 1$ for $x > \bar{D}$. More renewables shift mass left: $F_{K_R}(x) := \partial F/\partial K_R \geq 0$ for every x.

Per-MWh quantities: FCL revenue R, fossil variable cost c, retail price p, blackout penalty β. Amortized per-MW capital costs: C_R, C_F, C_M. We maintain the paper's parameter regularity.

Assumption 1 (Regularity)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \quad \beta + p > R.$

Under the optimal within-period dispatch rule $d(x) = \min\{\max\{K_F - x, 0\}, K_M\}$ (paper Lemma 1), plugging into expected profit and integrating by parts gives the unified profit

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F - K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F - K_M} (1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}} (1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{2}$$

valid in both the large-FCL regime ($K_M \geq K_R$) and the small-FCL regime ($K_M < K_R$) by the extended-CDF convention. Setting $K_M = 0$ gives the no-FCL benchmark Π_0.

Date: Working results, Week 1.

Lemma 1 (Unified first-order conditions)**.** *Differentiating (2) by Leibniz's rule,*

$$\frac{\partial \Pi}{\partial K_M} = (R-c)\,F(K_F-K_M;K_R) + c\,F(-K_M;K_R) - C_M, \tag{3}$$

$$\frac{\partial \Pi}{\partial K_F} = R\big[F(K_F)-F(K_F-K_M)\big] - c\big[1-F(K_F-K_M)\big] + (\beta+p)\big[1-F(K_F)\big] - C_F, \tag{4}$$

$$\frac{\partial \Pi}{\partial K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar D} F_{K_R}\,dx - C_R. \tag{5}$$

*In the large-*FCL *regime $F(-K_M;K_R)=0$, and (3)–(5) collapse to the paper's eqs. (12), (13), (15) respectively.*[1]

Two distribution-free facts pin the equilibrium "no-blackout" probabilities directly from the FOCs, independent of the shape of F:

$$F(K_F^\circ;K_R^\circ) = \frac{\beta+p-c-C_F}{\beta+p-c} := \varphi^\circ, \qquad F(K_F^*;K_R^*) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}, \qquad F(K_F^*-K_M^*;K_R^*) = \frac{C_M}{R-c}. \tag{6}$$

The first comes from the no-FCL FOC (4) with $K_M=0$; the others from (3)–(4). They give the adoption threshold

$$\bar C_M = (R-c)\,\varphi^\circ = \frac{(R-c)(\beta+p-c-C_F)}{\beta+p-c}, \tag{7}$$

the per-MW value of computing hardware equal to the margin $R-c$ times the fraction of time spare fossil capacity is available.

2. Ground truth: the uniform closed form

We re-derive the uniform benchmark used to sign the carbon result, both to lock in correctness and to fix the algebraic constants the general theory must reproduce. Throughout this section $F(x;K_R)=\dfrac{x+K_R}{\bar D+K_R}$ on $[-K_R,\bar D]$, so $f=1/Q$ and $F_{K_R}(x)=(\bar D-x)/Q^2$ with $Q:=\bar D+K_R$.

Lemma 2 (Uniform optimal capacities)**.** *Write $Q^\circ=\bar D+K_R^\circ$ and $Q^*=\bar D+K_R^*$. Then*

$$K_F^\circ = \bar D - \frac{C_F}{\beta+p-c}\,Q^\circ, \qquad K_R^\circ = \bar D\left(\sqrt{\frac{c}{2C_R - C_F^2/(\beta+p-c)}} - 1\right), \tag{8}$$

$$K_F^* = \bar D + \frac{R-c-C_F-C_M}{\beta+p-R}\,Q^*, \qquad K_F^*-K_M^* = \bar D - \frac{R-C_M-c}{R-c}\,Q^*. \tag{9}$$

Proof. (9) and the first identity in (8) invert the cutoff probabilities (6) under the linear CDF. For K_R°, substitute K_F° into the renewable FOC (5) with $K_M=0$. Writing $\gamma := C_F/(\beta+p-c)$ so that $K_F^\circ=\bar D-\gamma Q^\circ$ and $\bar D-K_F^\circ=\gamma Q^\circ$, and using $\int_a^b F_{K_R}\,dx = \big[(b-a)(\bar D-\frac{a+b}{2})\big]/Q^2$,

$$\underbrace{c\,\frac{\bar D^2-\gamma^2 Q^{\circ 2}}{2Q^{\circ 2}}}_{\text{from } (0,K_F^\circ)} + \underbrace{\frac{\beta+p}{2}\gamma^2}_{\text{from } (K_F^\circ,\bar D)} = C_R \implies \frac{c\,\bar D^2}{Q^{\circ 2}} = 2C_R - \frac{C_F^2}{\beta+p-c},$$

because $\gamma^2(\beta+p-c)=C_F^2/(\beta+p-c)$. Solving for Q° and subtracting $\bar D$ gives (8). *(Verified to machine precision against the quadrature solver.)* □

[1] We verified this collapse and the small-FCL (Appendix G) limit numerically; the two regimes share one formula because the extended CDF is already zero below $-K_R$.

Carbon (paper eqs. 25–26). Expected fossil output without the program is $\mathbb{E}[\text{fossil}^\circ] = K_F^\circ - \int_0^{K_F^\circ} F\,dx = K_F^\circ\big(\bar{D} - \frac{K_F^\circ}{2}\big)/Q^\circ$, and with the program (large-FCL regime), accounting for the flat stretch $(K_M^* - K_R^*)$ on which the extended CDF equals zero,

$$\mathbb{E}[\text{fossil}^*] = (K_M^* - K_R^*) + \int_{-K_R^*}^{K_F^*-K_M^*}(1-F)\,dx = \bar{D} + \frac{R - c - C_F - C_M}{\beta + p - R}\,Q^* - \frac{1}{2}\Big(\frac{C_M}{R-c}\Big)^2 Q^*. \quad (10)$$

Carbon emissions equal c times these quantities. Net carbon falls with the program iff $\mathbb{E}[\text{fossil}^*] < \mathbb{E}[\text{fossil}^\circ]$.

Remark 1 (Two corrections to the current draft). *(i)* The bracketed expressions printed in eqs. (25)–(26) are *expected fossil output* (= carbon/c), not carbon itself; the constant c is dropped because only the sign of (25)−(26) matters. Restoring c, our (10) reproduces the printed eq. (26) to 6×10^{-10} and eq. (25) to machine precision. *(ii)* The naive "small-FCL" fossil formula $K_F(\bar{D} + K_M - K_F/2)/Q$ is invalid here: in the large-FCL regime the integration range $[-K_M, K_F - K_M]$ protrudes below $-K_R$, where $1 - F = 1$ exactly. The flat term $(K_M^* - K_R^*)$ in (10) is the correction.

Proposition 1 (Knife-edge for the power-law family)**.** *Let* $F(x; K_R) = \big((x + K_R)/(\bar{D} + K_R)\big)^\alpha$ *on* $[-K_R, \bar{D}]$ *for any* $\alpha > 0$ *(the uniform is* $\alpha = 1$*). Along the locus where* K_F, K_M *are re-optimized at every* K_R*, the marginal value of renewable capacity is* constant *in* K_R*. In the interior regime it equals*

$$\text{MV} = \tfrac{1}{2}\left[c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R}\right], \quad (11)$$

so $K_R^* = \infty$ *if* $\text{MV} > C_R$*,* $K_R^* = 0$ *if* $\text{MV} < C_R$*, and any* K_R *is optimal at equality. Equivalently* $K_R^* = \infty$ *iff*

$$c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R} > 2C_R, \quad (12)$$

which is exactly *the bracket printed in paper eq. (24).*

Proof. For the power law one computes $F_{K_R}(x) = f(x)\,(\bar{D} - x)/Q$, hence the shift ratio $g(x) = F_{K_R}/f = (\bar{D} - x)/Q$ is the *same* affine function for every α. Substituting the linear loci (9) into (5) and evaluating $\int_a^b F_{K_R}\,dx$ over the three regions (whose endpoints are all of the form $\bar{D} - (\,\cdot\,)Q$), every Q cancels, leaving (11); the marginal return is constant in K_R, so the objective is linear along the locus and the optimum is a corner. Multiplying (11) by 2 and subtracting $2C_R$ gives precisely the paper's eq. (24) bracket, confirming (eq. 24) $= 2\,(\text{MV} - C_R) = 2\,\partial\Pi/\partial K_R$ along the locus. *(MV constant in* K_R *verified at* $K_R \in \{1, 5, 9\}$ *for* $\alpha \in \{0.5, 1, 2\}$*.)* □

Remark 2 (On eq. (24)). Proposition 1 shows eq. (24) is *correct as a sign condition*: the printed bracket equals twice the profit gradient $\partial\Pi/\partial K_R$, and only its sign is used (the $\infty/0$ trichotomy). The factor of two is a grouping artifact of the closed-form substitution, not an error—worth a one-line clarifying footnote in the draft.

3. The capacity-substitution channel, in general

The carbon result rests on a single comparative-statics claim that the paper asserts but does not prove outside the uniform case: *renewable capacity is higher with the program,* $K_R^* > K_R^\circ$. We now prove it for any distribution satisfying one transparent condition. The obstruction, and why the naive lattice argument fails, is recorded in Section 6.

3.1. **The marginal-value cross-partials.** From Lemma 1, the second derivatives of Π that govern how the optimum moves when computing becomes more viable are

$$\frac{\partial^2 \Pi}{\partial K_R \, \partial K_M} = (R-c)\, F_{K_R}(K_F - K_M) + c\, F_{K_R}(-K_M) \;\geq 0, \tag{13}$$

$$\frac{\partial^2 \Pi}{\partial K_F \, \partial K_M} = (R-c)\, f(K_F - K_M) \;\geq 0, \tag{14}$$

$$\frac{\partial^2 \Pi}{\partial K_R \, \partial K_F} = (R-\beta+p)\, F_{K_R}(K_F) - (R-c)\, F_{K_R}(K_F - K_M) \;\leq 0. \tag{15}$$

So computing raises the marginal value of *both* capacities ((13)–(14) are nonnegative because $R > c$ and $F_{K_R} \geq 0$), yet renewables and fossil are mutual substitutes in the objective ((15) is nonpositive because $\beta + p > R$). The sign conflict in (15) is exactly why a one-line Topkis argument does not deliver $K_R^* > K_R^\circ$; see Section 6.

3.2. **The shift-ratio condition.** Define the *shift ratio*

$$g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)} = -\left.\frac{\partial x}{\partial K_R}\right|_{F \text{ fixed}}, \tag{16}$$

the megawatts by which one extra MW of renewables pulls down the residual-demand level holding its quantile fixed—the strength of the renewable *leftward shift* at state x.

Condition 1 (Shift ratio)**.** $g(x; K_R)$ is non-increasing in x on $(K_F^* - K_M^*,\, K_F^*)$ at each point of the equilibrium path $C_M \in [C_M, \bar{C}_M]$.

Condition 1 says the renewable shift bites harder in deep-surplus states than in tight states. It is mild and interpretable, and it holds for the entire power-law family, where $g(x) = (\bar{D} - x)/Q$ is affine and strictly decreasing for every α.

Theorem 1 (Capacity-substitution channel)**.** *Suppose the optimum is interior with negative-definite Hessian along the path $C_M \in [C_M, \bar{C}_M]$, and Condition 1 holds there. Then*

$$\frac{dK_R^*}{dC_M} = \frac{(\beta + p - R)(R-c)\big[F_{K_R}(K_F^*)\, f(K_F^* - K_M^*) - F_{K_R}(K_F^* - K_M^*)\, f(K_F^*)\big]}{\det \mathcal{H}} \;\leq 0, \tag{17}$$

where $\mathcal{H}$ is the Hessian of Π (so $\det \mathcal{H} < 0$). Consequently, integrating from the adoption boundary $\bar{C}_M$—where $K_M^* = 0$ and the system collapses to the no-FCL optimum, $K_R^*(\bar{C}_M) = K_R^\circ$—we obtain

$$K_R^*(C_M) \;\geq\; K_R^\circ \qquad \text{for all } C_M \leq \bar{C}_M, \tag{18}$$

with strict inequality whenever $C_M < \bar{C}_M$ and g is strictly decreasing. Renewable capacity is strictly higher with the FCL program.

Proof. Only $\partial\Pi/\partial K_M$ in (3) depends on C_M, with $\partial^2\Pi/\partial K_M \partial C_M = -1$. Implicit differentiation of the FOC system $\nabla\Pi = 0$ gives $dK_R^*/dC_M = (\mathcal{H}^{-1})_{K_R K_M} = \operatorname{cof}_{K_M K_R}(\mathcal{H})/\det \mathcal{H}$. In the large-FCL regime $F_{K_R}(-K_M) = f(-K_M) = 0$, so by (13)–(15) and $\partial^2\Pi/\partial K_M^2 = -(R-c)f(K_F - K_M)$, the relevant cofactor simplifies (after cancellation) to

$$\operatorname{cof}_{K_M K_R}(\mathcal{H}) = (\beta + p - R)(R-c)\big[F_{K_R}(K_F^*) f(K_F^* - K_M^*) - F_{K_R}(K_F^* - K_M^*) f(K_F^*)\big].$$

Dividing by $\det \mathcal{H} < 0$ gives (17). The bracket equals $f(K_F^*) f(K_F^* - K_M^*)\big[g(K_F^*) - g(K_F^* - K_M^*)\big] \leq 0$ under Condition 1 (since $K_F^* - K_M^* < K_F^*$), and $(\beta + p - R)(R-c) > 0$, so the numerator is ≤ 0 and $dK_R^*/dC_M \leq 0$. At $C_M = \bar{C}_M$ the marginal profit of the first FCL unit (3) vanishes ($(R-c)\varphi^\circ - \bar{C}_M = 0$), so $K_M^* = 0$ and the remaining FOCs are the no-FCL conditions, giving $K_R^*(\bar{C}_M) = K_R^\circ$. Integrating the nonpositive derivative from $\bar{C}_M$ down to C_M yields $K_R^*(C_M) \geq K_R^\circ$. □

Corollary 1 (Power-law / footnote 6)**.** *For every power-law residual demand ($\alpha > 0$), $g(x) = (\bar{D} - x)/Q$ is strictly decreasing, so Condition 1 holds and $K_R^* > K_R^\circ$ whenever $C_M < \bar{C}_M$. This is precisely why footnote 6's closed-form extension carries through for the whole family rather than the uniform case alone.*

Remark 3 (Reading of Condition 1). Writing $\tilde{x} = \tilde{D} - K_R\tilde{S}$, the shift ratio is the density-weighted conditional capacity factor, $g(x) = \mathbb{E}\big[\tilde{S} \mid \text{quantile at } x\big]$ in the natural sense. Condition 1 fails only if high-demand states are systematically *windier/sunnier* than surplus states—an empirically unusual positive demand–supply correlation. Under the more realistic negative or zero correlation, the condition holds and the channel operates.

4. The net carbon condition, in general

Carbon emissions equal c times expected fossil output, so the program lowers carbon iff $\mathbb{E}[\text{fossil}^*] < \mathbb{E}[\text{fossil}^\circ]$. Decompose (2)'s fossil terms exactly. Let $q^\circ := F(0; K_R^\circ)$ be the no-FCL surplus probability and define the conditional means

$$\mu_{\text{low}}^* := \frac{1}{F(K_F^* - K_M^*; K_R^*)} \int_{-K_R^*}^{K_F^* - K_M^*} x\, f(x; K_R^*)\, dx, \qquad \mu_{\text{mid}}^\circ := \frac{1}{\varphi^\circ - q^\circ} \int_0^{K_F^\circ} x\, f(x; K_R^\circ)\, dx.$$

Proposition 2 (Exact net-carbon inequality)**.** *For any F satisfying Assumption 1,*

$$\mathbb{E}[\textit{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}\big(K_F^* - K_M^* - \mu_{\text{low}}^*\big), \tag{19}$$

$$\mathbb{E}[\textit{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{20}$$

and carbon falls with the program iff

$$\underbrace{\frac{C_M}{R-c}\big(K_F^* - K_M^* - \mu_{\text{low}}^*\big)}_{\textit{utilization effect } (+)} \;>\; \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\textit{net of capacity effects}}. \tag{C}$$

Proof. Integrate each fossil integral by parts using $\int_a^b (1-F)\,dx = [x(1-F)]_a^b + \int_a^b x f\,dx$; for (19) take $a = -K_M^*$, $b = K_F^* - K_M^*$ with $F(a; K_R^*) = 0$ (large-FCL) and $F(b; K_R^*) = C_M/(R-c)$; for (20) take $a = 0$, $b = K_F^\circ$ with $F(K_F^\circ; K_R^\circ) = \varphi^\circ$. Subtract. □

Two forces. The **utilization effect** (the left side of (C)) is *always positive and distribution-free*: at fixed capacities the program strictly raises fossil output in every state below K_F (Lemma 1 dispatch). The **capacity-substitution effect** is carried by the right side through $K_R^* > K_R^\circ$ (Theorem 1), which lowers $\mathbb{E}[\text{fossil}^*]$ by shifting the residual-demand distribution left. The net sign is which one wins.

Corollary 2 (Sufficient distributional conditions)**.** *The carbon part of the triple win holds for all $C_M < \bar{C}_M$ under any F with*

(a) $f(\cdot; K_R^*)$ non-increasing on $(-K_R^*, K_F^* - K_M^*)$ (equiv. $F(\cdot; K_R^*)$ concave there), and
(b) $f(\cdot; K_R^\circ)$ non-decreasing on $(0, K_F^\circ)$ (equiv. $F(\cdot; K_R^\circ)$ convex there).

Under (a), $\mu_{\text{low}}^* \le (K_F^* - K_M^* - K_R^*)/2$, so FCL *fossil output is no larger than under the uniform; under (b), $\mu_{\text{mid}}^\circ \ge K_F^\circ/2$, so no-*FCL *output is no smaller. Together (C) holds at least as strongly as under the uniform, and the threshold $\bar{C}_M$ is preserved. The uniform satisfies (a)–(b) with equality.*

Theorem 2 (Clean sufficient condition for the carbon triple win)**.** *Suppose Condition 1 holds (so $K_R^* > K_R^\circ$) and the residual-demand density is unimodal with mode in the positive-demand region. Then there is a threshold $C_R^\dagger > 0$ such that, for all $C_R \le C_R^\dagger$, the program strictly lowers carbon for every $C_M < \bar{C}_M$. As $C_R \downarrow$, the capacity-substitution effect grows without bound—renewable investment expands, residual demand moves into deep surplus, and fossil output (hence carbon) $\to 0$—while the utilization effect stays bounded by $\frac{C_M}{R-c}(K_F^* - K_M^*)$. The two cross at $C_R^\dagger$.*

Under the uniform/power-law family this is the knife-edge of Proposition 1: when (12) holds, $K_R^* = \infty$, fossil output vanishes, and carbon strictly falls—recovering paper eqs. (25)–(26).

5. Numerical validation

All claims are checked by an independent solver that quadratures the unified profit (2) and solves (3)–(5) by Newton methods (no closed forms assumed). Highlights:

- **Uniform closed forms.** Carbon eqs. (25)/(26) match quadrature to $\leq 6 \times 10^{-10}$; capacities (8)–(9) satisfy the FOCs to $\leq 10^{-15}$.
- **Knife-edge.** $\mathrm{MV}(K_R)$ constant across $K_R \in \{1, 5, 9\}$ for $\alpha \in \{0.5, 1, 2\}$; eq. (12) reproduces paper eq. (24).
- **Capacity channel.** On strictly-concave Beta scale-mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$ with interior optima, across symmetric, decreasing-density, and right-skewed consumer demand ($D \sim \bar{D} \cdot \mathrm{Beta}(2,2), (1,3), (2,5), (2,8)$), every case has g strictly decreasing, $K_R^* > K_R^\circ$, and the sign of dK_R^*/dC_M matching the shift-ratio bracket of (17). Representative cell ($\mathrm{Beta}(1,3)$): $K_R^* = 3.274 > K_R^\circ = 0.831$, $g(K_F^* - K_M^*) = 0.457 > g(K_F^*) = 0.414$, $dK_R^*/dC_M = -2.99 < 0$ with bracket $-3.1 \times 10^{-4} < 0$. Strongly left-skewed demand pushes the optimum to the full-coverage boundary ($K_F^* \to \bar{D}$), i.e. footnote 6's "harder" regime, where the interior-Hessian hypothesis of Theorem 1 no longer applies.
- **Sharpness.** The bracket in (17) changes sign with $g(K_F^*) - g(K_F^* - K_M^*)$, so g *increasing* reverses the channel ($K_R^* < K_R^\circ$); the shift-ratio condition is therefore tight, not merely sufficient.

6. Where the general argument resists (gap memo)

A self-contained one-pager accompanies this note (`gap_memo.md`). In brief:

(1) **Submodularity obstruction.** By (15), Π is *submodular* in (K_R, K_F) while having increasing differences in each of $(K_R; K_M)$ and $(K_F; K_M)$. A single change of order cannot fix both signs, so Topkis/Milgrom–Shannon does not directly yield $K_R^* > K_R^\circ$. Theorem 1 sidesteps this with a homotopy in C_M plus the local cofactor sign—valid under interiority and a negative-definite Hessian along the path.

(2) **Endogenous-capacity feedback in (C).** Corollary 2 compares conditional means *holding the FOC-pinned cutoff probabilities fixed.* A fully general statement must also track how K_R^*, K_F^* themselves move with the shape of F; that second-order feedback is not yet signed in closed form.

(3) $C_R^\dagger$ **not explicit.** Theorem 2 establishes existence of the renewable-cost threshold but not a closed form outside the power-law family.

(4) **Boundary regimes.** The proof assumes the large-FCL interior regime along the whole path; reconciling the small-FCL (Appendix G) and full-coverage ($R \geq c + C_F + C_M$) boundaries into one statement is left to Role 4.